

Skill: GIT

File Name: 3.GIT-HOL

Git Branch Creation and Status Check

```
MINGW64/c/Users/hp/GitDemo

hp@Geethika MINGW64 ~
$ cd ~/GitDemo

hp@Geethika MINGW64 ~/GitDemo (master)
$ git branch GitNewBranch

hp@Geethika MINGW64 ~/GitDemo (master)
$ git branch -a
  GitNewBranch
* master
  remotes/origin/master

hp@Geethika MINGW64 ~/GitDemo (master)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'

hp@Geethika MINGW64 ~/GitDemo (GitNewBranch)
$ echo "This content is from GitNewBranch" > feature.txt

hp@Geethika MINGW64 ~/GitDemo (GitNewBranch)
$ git add feature.txt
git commit -m "Add feature.txt in GitNewBranch"
warning: in the working copy of 'feature.txt', LF will be replaced by CRLF the next time Git touches it
[GitNewBranch 4c1f60b] Add feature.txt in GitNewBranch
 1 file changed, 1 insertion(+)
 create mode 100644 feature.txt

hp@Geethika MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

hp@Geethika MINGW64 ~/GitDemo (GitNewBranch)
$ |
```

CLI Code Differences Between Branches

```
MINGW64/c/Users/hp/GitDemo

hp@Geethika MINGW64 ~
$ cd ~/GitDemo

hp@Geethika MINGW64 ~/GitDemo (master)
$ git branch GitNewBranch

hp@Geethika MINGW64 ~/GitDemo (master)
$ git branch -a
* master
  remotes/origin/master

hp@Geethika MINGW64 ~/GitDemo (master)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'

hp@Geethika MINGW64 ~/GitDemo (GitNewBranch)
$ echo "This content is from GitNewBranch" > feature.txt

hp@Geethika MINGW64 ~/GitDemo (GitNewBranch)
$ git add feature.txt
git commit -m "Add feature.txt in GitNewBranch"
warning: in the working copy of 'feature.txt', LF will be replaced by CRLF the next time Git touches it
[GitNewBranch 4clf60b] Add feature.txt in GitNewBranch
1 file changed, 1 insertion(+)
 create mode 100644 feature.txt

hp@Geethika MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

hp@Geethika MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

hp@Geethika MINGW64 ~/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/feature.txt b/feature.txt
new file mode 100644
index 0000000..a193630
--- /dev/null
+++ b/feature.txt
@@ -0,0 +1 @@
+This content is from GitNewBranch

hp@Geethika MINGW64 ~/GitDemo (master)
$
```

Local Git Branch Merge and Graph Logging

```
hp@Geethika MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch
Updating b70f066..4clf60b
Fast-forward
 feature.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 feature.txt

hp@Geethika MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 4clf60b (HEAD -> master, GitNewBranch) Add feature.txt in GitNewBranch
* b70f066 (origin/master) Add .gitignore to exclude log files and folders
* a91c44c Initial commit for the GitDemo project

hp@Geethika MINGW64 ~/GitDemo (master)
$
```

Configuring External Difference Tool

```
hp@Geethika MINGW64 ~/GitDemo (master)
$ git config --global diff.tool p4merge

hp@Geethika MINGW64 ~/GitDemo (master)
$ git config --global difftool.p4merge.path "C:/Program Files/Perforce/p4merge.exe"

hp@Geethika MINGW64 ~/GitDemo (master)
$ git config --global merge.tool p4merge

hp@Geethika MINGW64 ~/GitDemo (master)
$ git config --global mergetool.p4merge.path "C:/Program Files/Perforce/p4merge.exe"

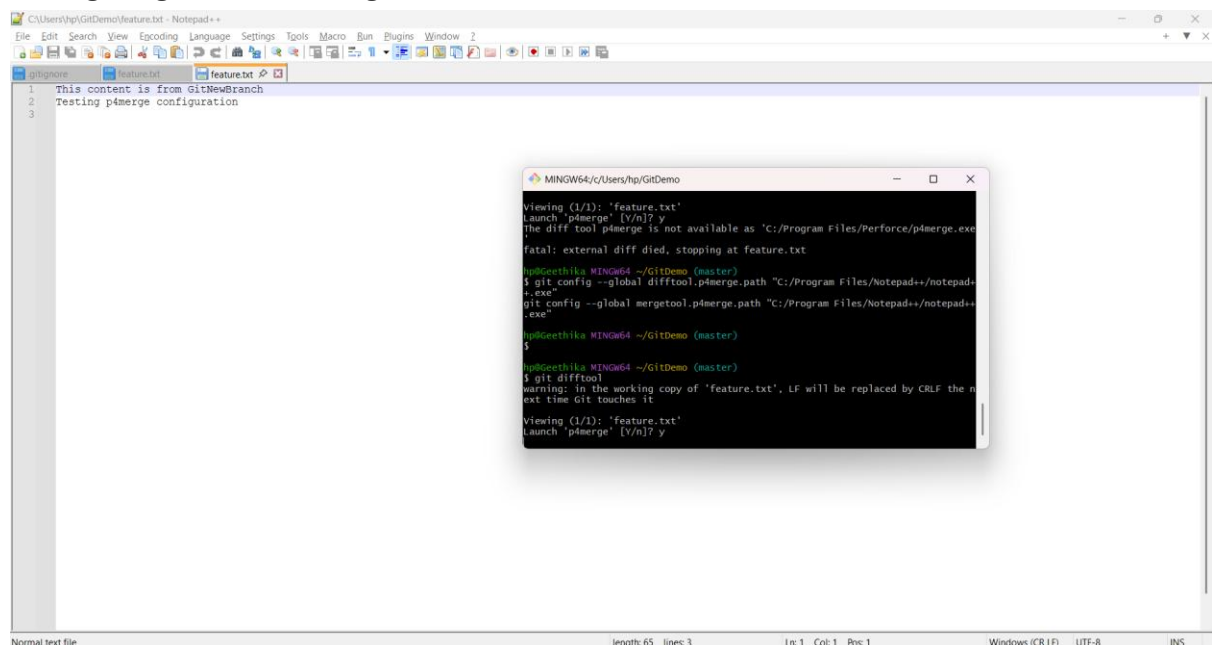
hp@Geethika MINGW64 ~/GitDemo (master)
$ echo "Testing p4merge configuration" >> feature.txt

hp@Geethika MINGW64 ~/GitDemo (master)
$ git difftool
warning: in the working copy of 'feature.txt', LF will be replaced by CRLF the next time Git touches it

Viewing (1/1): 'feature.txt'
Launch 'p4merge' [Y/n]? y
The diff tool p4merge is not available as 'C:/Program Files/Perforce/p4merge.exe'
fatal: external diff died, stopping at feature.txt

hp@Geethika MINGW64 ~/GitDemo (master)
$
```

Configuring and Launching External Difference Tool



Final Workspace Verification and Clean Status

```
hp@Geethika MINGW64 ~/GitDemo (master)
$ git checkout feature.txt
Updated 1 path from the index

hp@Geethika MINGW64 ~/GitDemo (master)
$
```